

Cloud Computing

- The term cloud refers to a **network or the internet**. It is a technology that uses remote servers on the internet to store, manage, and access data online rather than local drives. The data can be anything such as files, images, documents, audio, video, and more.

There are the following operations that we can do using cloud computing:

- Developing new applications and services
- Storage, back up, and recovery of data
- Hosting blogs and websites
- Delivery of software on demand
- Analysis of data
- Streaming videos and audios

Characteristics of Cloud Computing

1) Agility

The cloud **works in a distributed computing environment**. It shares resources among users and works very fast.

2) High availability and reliability

The availability of servers is high and more reliable because the **chances of infrastructure failure are minimum**.

3) High Scalability

Cloud offers "**on-demand**" provisioning of resources on a large scale, without having engineers for peak loads.

4) Multi-Sharing

With the help of cloud computing, **multiple users and applications can work more efficiently** with cost reductions by sharing common infrastructure.

5) Device and Location Independence

Cloud computing enables the users to access systems using a web browser regardless of their location or what device they use e.g. PC, mobile phone, etc. **As infrastructure is off-site** (typically provided by a third-party) **and accessed via the Internet, users can connect from anywhere.**

6) Maintenance

Maintenance of cloud computing applications is easier, since they **do not need to be installed on each user's computer and can be accessed from different places.** So, it reduces the cost also.

7) Low Cost

By using cloud computing, the cost will be reduced because to take the services of cloud computing, **IT company need not to set its own infrastructure** and pay-as-per usage of resources.

8) Services in the pay-per-use mode

Application Programming Interfaces (APIs) **are provided to the users so that they can access services on the cloud** by using these APIs **and pay the charges as per the usage of services.**

Advantages of Cloud Computing

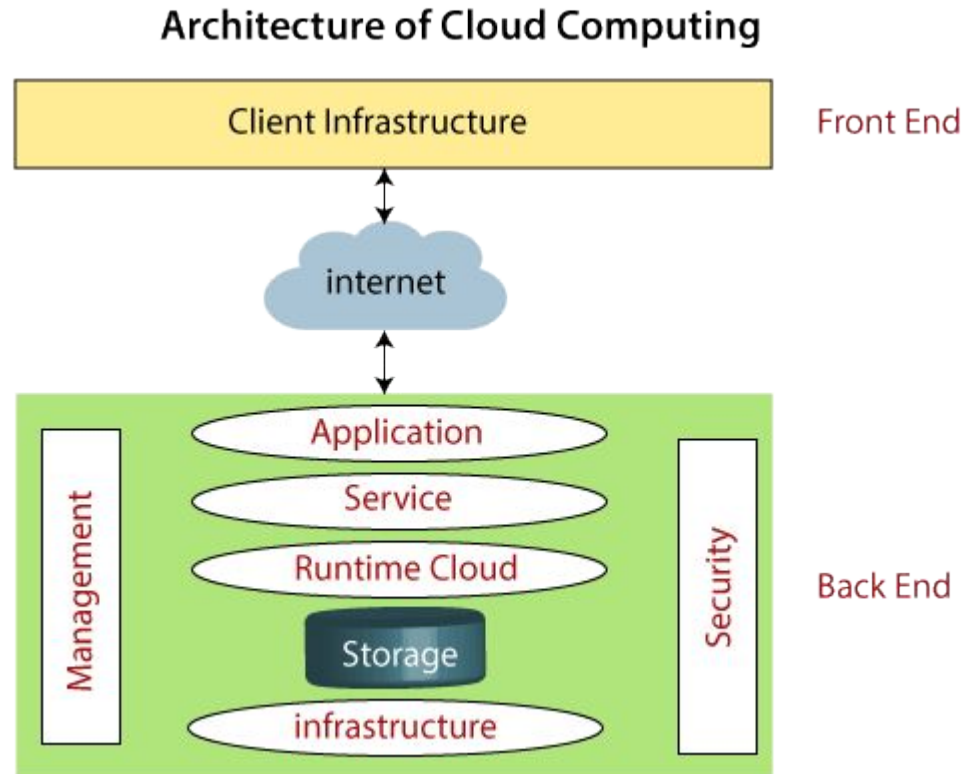
1) Advantages

- Back-up and restore data
- Improved collaboration
- Excellent accessibility
- Low maintenance cost
- Data security

2) Disadvantages

- Internet Connectivity
- Vendor lock-in
- Limited Control

Cloud Computing Architecture



Cloud Computing Architecture Components

1. Client Infrastructure

Client Infrastructure is a Front end component. It provides GUI (Graphical User Interface) to interact with the cloud.

2. Application

The application may be any software or platform that a client wants to access.

3. Service

A Cloud Services manages that which type of service you access according to the client's requirement.

4. Runtime Cloud

Runtime Cloud provides the **execution and runtime environment** to the virtual machines.

5. Storage

Storage is one of the most important components of cloud computing. It provides a huge amount of storage capacity in the cloud to store and manage data.

Cloud Computing Architecture Components

6. Infrastructure

It provides services on the **host level**, **application level**, and **network level**. Cloud infrastructure includes hardware and software components such as servers, storage, network devices, virtualization software, and other storage resources that are needed to support the cloud computing model.

7. Management

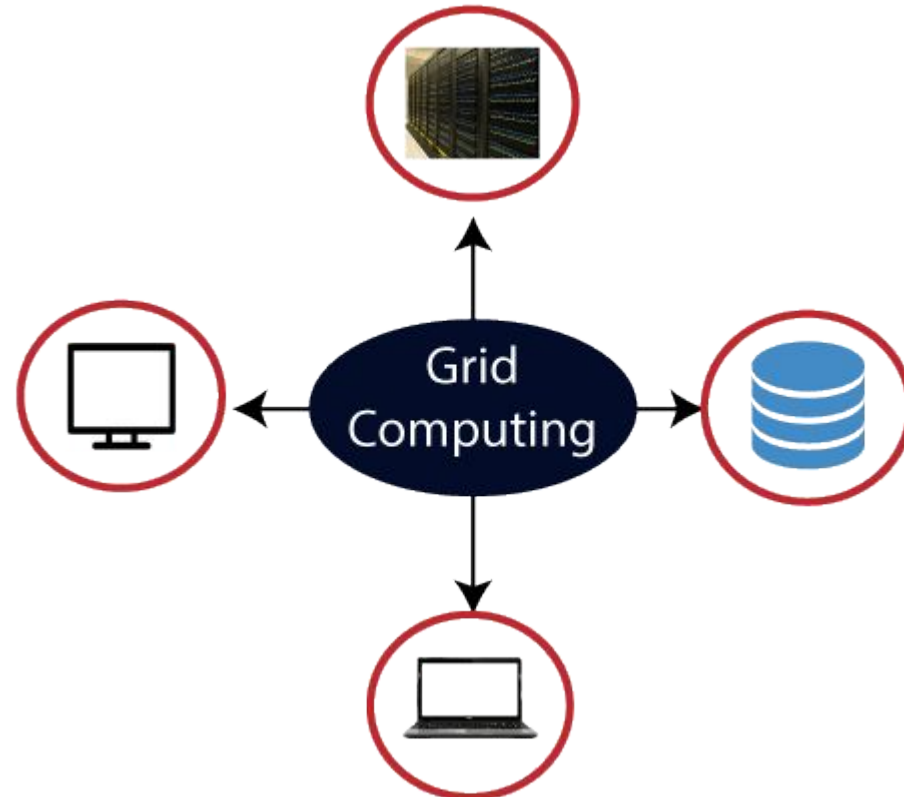
Management is used to manage components such as application, service, runtime cloud, storage, infrastructure, and other security issues in the backend and establish coordination between them.

8. Security

Security is an in-built back end component of cloud computing. It implements a security mechanism in the back end.

Grid Computing

- Grid computing is also called as "distributed computing." It links multiple computing resources (PC's, workstations, servers, and storage elements) together and provides a mechanism to access them.
- The main advantages of grid computing are that it increases user productivity by providing transparent access to resources, and work can be completed more quickly.



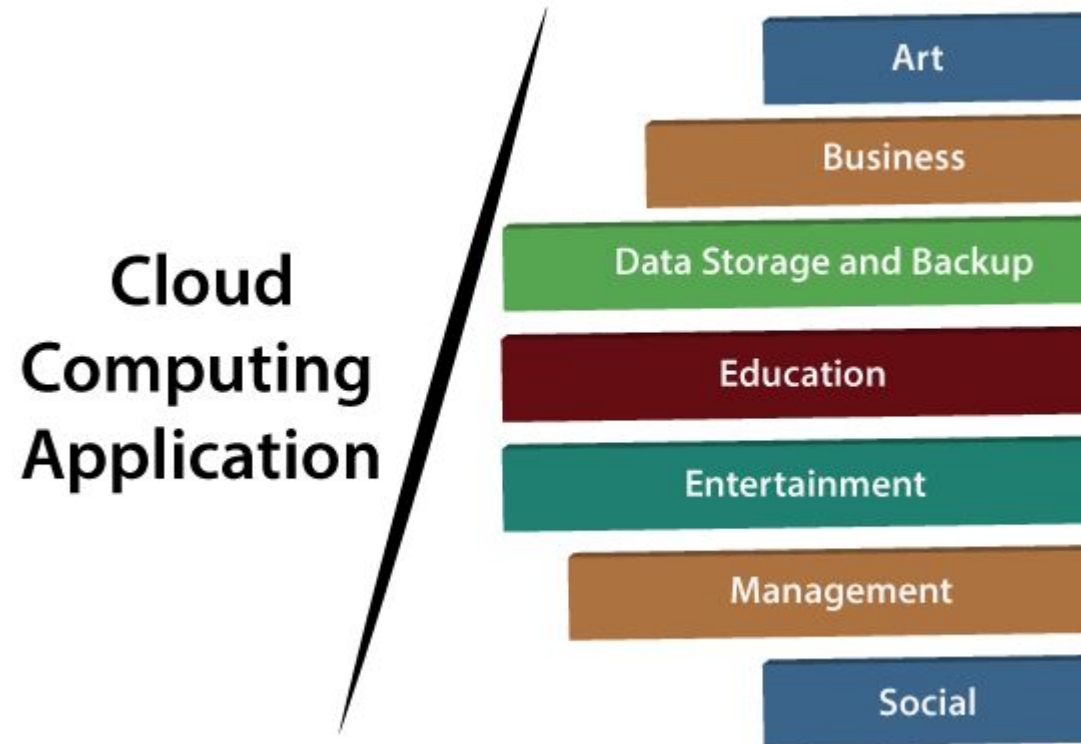
Grid Computing

- It refers to a distributed computing approach that involves the coordination and sharing of computing resources, such as processing power, storage, and data, across a network of interconnected computers.
- In grid computing, multiple computers are connected to form a virtual supercomputer, and these computers work together to solve complex computational tasks or process large datasets. It's often used for scientific research, data-intensive calculations, and simulations. Each computer in the grid contributes its unused processing power and resources to the collective effort, creating a powerful and scalable computing environment.
- While grid computing can involve network communication and access to servers, its primary purpose is to harness distributed computing resources for parallel processing and problem-solving.
- Grid computing is more about achieving high-performance computing and solving complex problems efficiently by leveraging a network of computers.

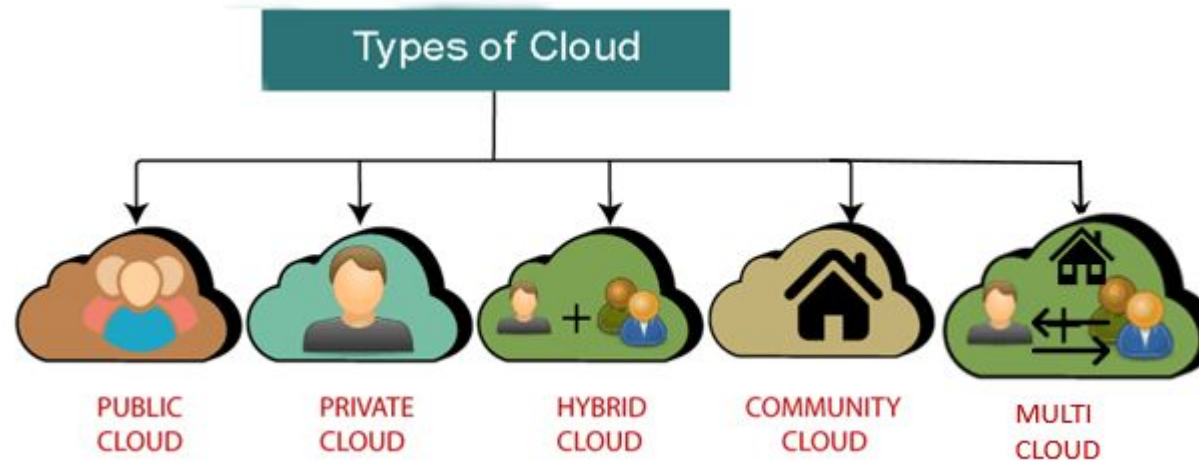
Cloud Computing **VS** Grid Computing

Cloud Computing	Grid Computing
Cloud Computing follows client-server computing architecture.	Grid computing follows a distributed computing architecture.
Scalability is high.	Scalability is normal.
Cloud Computing is more flexible than grid computing.	Grid Computing is less flexible than cloud computing.
Cloud operates as a centralized management system.	Grid operates as a decentralized management system.
In cloud computing, cloud servers are owned by infrastructure providers.	In Grid computing, grids are owned and managed by the organization.
Cloud computing uses services like Iaas, PaaS, and SaaS.	Grid computing uses systems like distributed computing, distributed information, and distributed pervasive.
Cloud Computing is Service-oriented.	Grid Computing is Application-oriented.
It is accessible through standard web protocols.	It is accessible through grid middleware.

Cloud Computing Applications



Types of Cloud



1. **Public cloud** is open to all to store and access information via the Internet using the pay-per-usage method. In public cloud, computing resources are managed and operated by the Cloud Service Provider (CSP). The CSP looks after the supporting infrastructure and ensures that the resources are accessible to and scalable for the users.
2. **Private cloud** is also known as an internal cloud or corporate cloud. It is used by organizations to build and manage their own data centers internally or by the third party. It can be deployed using Opensource tools such as Openstack and Eucalyptus.

Types of Cloud

3. **Hybrid Cloud** is a combination of the public cloud and the private cloud.

4. **Community cloud** allows systems and services to be accessible by a group of several organizations to share the information between the organization and a specific community. It is owned, managed, and operated by one or more organizations in the community, a third party, or a combination of them.

5. **Multi-cloud** is a strategy in cloud computing where companies utilize more than one cloud service provider or platform to meet their computing needs. It involves distributing workloads, applications, and statistics throughout numerous cloud environments consisting of public, private, and hybrid clouds.

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Cloud Service Model

1. **IaaS** is also known as Hardware as a Service (HaaS). It is a computing infrastructure managed over the internet. The main advantage of using IaaS is that it helps users to avoid the cost and complexity of purchasing and managing the physical servers.
Example: DigitalOcean, Linode, Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE), Rackspace, and Cisco Metacloud.
2. **PaaS** cloud computing platform is created for the programmer to develop, test, run, and manage the applications.
Example: AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, Magento Commerce Cloud, and OpenShift.
3. **SaaS** is also known as "on-demand software". It is a software in which the applications are hosted by a cloud service provider. Users can access these applications with the help of internet connection and web browser.
Example: BigCommerce, Google Apps, Salesforce, Dropbox, ZenDesk, Cisco WebEx, ZenDesk, Slack, and GoToMeeting.

Cloud Service Model

IaaS	Paas	SaaS
It provides a virtual data center to store information and create platforms for app development, testing, and deployment.	It provides virtual platforms and tools to create, test, and deploy apps.	It provides web software and apps to complete business tasks.
It provides access to resources such as virtual machines, virtual storage, etc.	It provides runtime environments and deployment tools for applications.	It provides software as a service to the end-users.
It is used by network architects.	It is used by developers.	It is used by end users.
IaaS provides only Infrastructure.	PaaS provides Infrastructure+Platform.	SaaS provides Infrastructure+Platform+Software.